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Heat transport device

Abstract

The heat transport device includes a first wick structure provided in the internal space of the first heat transport section, and extending from the evaporation section to the condensation section, a second wick structure provided on an inner surface of the second heat transport section, and extending along the heat transport direction of the second heat transport section, and a reflux promoting body provided on the second wick structure, extending along the heat transport direction of the second heat transport section, and having a capillary force, wherein a capillary force of the first wick structure is larger than the capillary force of the reflux promoting body, and the capillary force of the reflux promoting body is larger than a capillary force of the second wick structure, or the capillary force of the reflux promoting body is equal to the capillary force of the second wick structure.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4116266	12/1977	Sawata	138/40	F28D 15/046
4118756	12/1977	Nelson	165/104.33	H05K 1/0272
4351388	12/1981	Calhoun	165/104.26	F28D 15/046
11747092	12/2022	Chen	165/104.26	F28D 15/0233
11913725	12/2023	Cheng	N/A	F28D 15/0233
2007/0240858	12/2006	Hou et al.	N/A	N/A
2016/0014931	12/2015	Hamakawa	N/A	N/A
2020/0018555	12/2019	Lin	N/A	N/A
2020/0103175	12/2019	Ben-Menahem et al.	N/A	N/A
2020/0173730	12/2019	Watanabe et al.	N/A	N/A
2020/0393201	12/2019	Watanabe et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
101055158	12/2006	CN	N/A
205092233	12/2015	CN	N/A
110785071	12/2019	CN	N/A
212876430	12/2020	CN	N/A
3686540	12/2019	EP	N/A
3758057	12/2019	EP	N/A
3857152	12/2020	EP	N/A

3-179766	12/1990	JP	N/A
2015-121373	12/2014	JP	N/A
2016-009828	12/2015	JP	N/A
6582114	12/2018	JP	F28D 15/0233
2020-85430	12/2019	JP	N/A
2020-176752	12/2019	JP	N/A
201447215	12/2013	TW	N/A
202028679	12/2019	TW	N/A
202040775	12/2019	TW	N/A
2014/157147	12/2013	WO	N/A
2020/068212	12/2019	WO	N/A
2020/110973	12/2019	WO	N/A
WO-2020213581	12/2019	WO	F28D 15/0233

OTHER PUBLICATIONS

International Search Report and Written Opinion dated Feb. 14, 2023 from corresponding International Patent Application No. PCT/JP2022/048554, 11 pages. cited by applicant
International Preliminary Report on Patentability dated Sep. 6, 2024 from corresponding International Patent Application No. PCT/JP2022/048554, 11 pages. cited by applicant
Japanese Office Action dated Apr. 7, 2022 from Corresponding Japanese Patent Application No. 2022-027148, 12 pages. cited by applicant

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation application of International Patent Application No. PCT/JP2022/048554 filed on Dec. 28, 2022, which claims the benefit of Japanese Patent Application No. 2022-027148, filed on Feb. 24, 2022. The contents of these applications are incorporated herein by reference in their entirety.

BACKGROUND

Technical Field

(1) The present disclosure relates to a heat transport device that cools heating elements such as electric/electronic components or the like mounted on a substrate such as an electronic circuit board, and particularly relates to a heat transport device that can prevent a working fluid in an evaporation section from drying out even under usage conditions in which a heating element with a high heat generation amount is thermally connected, or in usage conditions in which a flow rate of cooling air is increased.

Background

(2) With high functionality of electronic equipment, many components including heating elements such as electronic components are mounted on a substrate such as an electronic circuit board, inside of the electronic equipment. Further, with high functionality of electronic equipment, the heat generation amounts of the heating elements such as electronic components are increasing. As a unit configured to cool the heating elements such as electronic components, heat transport members such as a vapor chamber, a heat pipe, and a heat sink may be used.

(3) However, when the heat generation amount of a heating element such as an electronic

component increases, the flow rate of the working fluid in a gas phase that flows to a condensation section from an evaporation section of the heat transport member increases, so that a reflux of the working fluid in a liquid phase that is in a counterflow relationship with the working fluid in a gas phase has sometimes been hindered. In particular, when the capillary force of the wick structure is relatively small, such as in the case of the wick structure including a plurality of groove portions, the reflux of the working fluid tends to be hindered. When the reflux of the working fluid in a liquid phase is hindered, the working fluid in a liquid phase accumulates in the condensation section of the heat transport member, as a result of which, a reflux amount of the working fluid in a liquid phase to the evaporation section from the condensation section may be insufficient, and the heat transport member may dry out.

(4) Thus, as the heat transport member that cools the heating elements such as electronic components, there is proposed a vapor chamber in which a wick structure having a relatively large capillary force such as a sintered body of metal powder, and the working fluid are housed, which is a vapor chamber including a first section, a second section extending from one end of the first section in a direction to be away from the first section, a fourth section extending from the second section in a left-to-right direction, a third section extending from one end of the first section in a direction to be away from the first section, and a fifth section extending from the third section in a left-to-right direction, in which the wick structure extends throughout all of the first section to the fifth section (Specification of U. S. Patent Application Publication No. 2020/0018555).

(5) In U. S. Patent Application Publication No. 2020/0018555, the heating elements is cooled by transporting heat of the heating element thermally connected to the first section to the fourth section and the fifth section that are condensation sections from the first section that is an evaporation section via the second section and the third section. Further, by the wick structure having a relatively large capillary force such as a sintered body of metal powder, the working fluid in a liquid phase is caused to flow back to the first section from the fourth section and the fifth section.

(6) However, in U. S. Patent Application Publication No. 2020/0018555, the same wick structure extends throughout all of the first section to the fifth section, and therefore the wick structure positioned in the fourth section and the fifth section tends to retain the working fluid in a liquid phase. Accordingly, also in the vapor chamber of U. S. Patent Application Publication No. 2020/0018555, the working fluid in a liquid phase still accumulates in the condensation section, a reflux amount of the working fluid in a liquid phase to the evaporation section from the condensation section may be insufficient, and the working fluid may dry out.

(7) Further, in order to smoothly cool the heating element with a high heat generation amount, a flow rate of cooling air that is to be supplied to the heat transport member may be increased. However, if the flow rate of the cooling air is increased, phase change of the working fluid from a gas phase to a liquid phase is promoted in the condensation section, and in the vapor chamber of U. S. Patent Application Publication No. 2020/0018555, the working fluid in a liquid phase still accumulates in the condensation section, and may dry out.

SUMMARY

(8) The present disclosure is related to providing a heat transport device that can prevent a working fluid in an evaporation section from drying out even under usage conditions in which a heating element with a high heat generation amount is thermally connected, or usage conditions in which a flow rate of cooling air is increased.

(9) The gist of a configuration of a heat sink of the present disclosure is as follows.

(10) [1] A heat transport device including

(11) a first heat transport section including an evaporation section thermally connected to a heating element, and a second heat transport section connected at a condensation section of the first heat transport section, and having a heat transport direction different from a heat transport direction of the first heat transport section, the first heat transport section including an integral internal space

that provides communication from the evaporation section to a connection portion with the second heat transport section and has a working fluid enclosed in the internal space, the internal space of the first heat transport section communicating with an internal space of the second heat transport section, the heat transport device including a first wick structure provided in the internal space of the first heat transport section, and extending from the evaporation section to the condensation section, a second wick structure provided on an inner surface of the second heat transport section, and extending along the heat transport direction of the second heat transport section, and a reflux promoting body provided on the second wick structure, extending along the heat transport direction of the second heat transport section, and having a capillary force, wherein a capillary force of the first wick structure is larger than the capillary force of the reflux promoting body, and the capillary force of the reflux promoting body is larger than a capillary force of the second wick structure, or the capillary force of the reflux promoting body is equal to the capillary force of the second wick structure.

[2] The heat transport device as set forth in [1], wherein the first wick structure and the reflux promoting body are connected via a connection member having a capillary force.

[3] The heat transport device as set forth in [2], wherein the capillary force of the first wick structure is larger than the capillary force of the connection member, and the capillary force of the connection member is larger than the capillary force of the reflux promoting body or the capillary force of the connection member is equal to the capillary force of the reflux promoting body.

[4] The heat transport device as set forth in [2] or [3], wherein the connection member is positioned at the connection portion of the first heat transport section and the second heat transport section.

[5] The heat transport device as set forth in any one of [1] to [4], wherein in a section in an orthogonal direction to the heat transport direction of the second heat transport section, the reflux promoting body covers a region of 5% or more of a peripheral length of the section.

[6] The heat transport device as set forth in any one of [1] to [5], wherein the reflux promoting body includes a plurality of reflux promoting units, and capillary forces of the plurality of reflux promoting units are different from one another.

[7] The heat transport device as set forth in any one of [1] to [6], wherein a container of the first heat transport section is of a planar type.

[8] The heat transport device as set forth in any one of [1] to [7], wherein a container of the second heat transport section is of a planar type including an integral internal space.

[9] The heat transport device as set forth in any one of [1] to [7], wherein the second heat transport section is a plurality of tube bodies.

[10] The heat transport device as set forth in any one of [1] to [9], wherein a plurality of the second heat transport sections are provided, and extend in a plurality of directions from the first heat transport section.

[11] The heat transport device as set forth in [9], wherein the plurality of tube bodies extend downward in a gravity direction.

[12] The heat transport device as set forth in any one of [1] to [11], wherein the heat transport device is a heat sink in which a plurality of heat dissipation fins are thermally connected to the second heat transport section.

(12) In an aspect of the heat transport device of the present disclosure, the internal space of the heat transport member including the evaporation section (that is, a heat receiving section) is integrated with communication being provided in a whole of the internal space unlike an internal space of a heat pipe group in which a plurality of heat pipes are disposed in parallel. In the aspect of the heat transport device of the present disclosure, in the first heat transport section, the part that is thermally connected to the heating element that is the object to be cooled functions as the evaporation section (that is, the heat receiving section), and the part that is thermally connected to the second heat transport section functions as the condensation section (that is, a heat dissipation section). In the evaporation section of the first heat transport section, the working fluid receives

heat from the heating element and changes in phase from a liquid phase to a gas phase, and in the condensation section of the first heat transport section, part of the working fluid in a gas phase releases latent heat to change in phase from a gas phase to a liquid phase. In the aspect of the heat transport device of the present disclosure, the heat of the heating element is transported to the condensation section of the first heat transport section from the evaporation section of the first heat transport section by the first heat transport section, and is further transported to the second heat transport section from the condensation section of the first heat transport section. Further, the working fluid changing in phase to a gas phase by the first heat transport section receiving heat from the heating element flows to the second heat transport section from the first heat transport section. The working fluid in a gas phase flows to the second heat transport section from the first heat transport section, whereby the second heat transport section receives heat from the first heat transport section, and the heat of the heating element diffuses to the second heat transport section from the first heat transport section. The heat exchange section such as the heat dissipating fins are thermally connected to the second heat transport section, whereby the second heat transport section further transmits the heat received from the first heat transport section to the heat exchange section, and the heat is released to the external environment from the heat exchange section. When the second heat transport section transmits the heat received from the first heat transport section to the heat exchange section, the working fluid in a gas phase that flows to the second heat transport section from the first heat transport section changes in phase to a liquid phase.

(13) Since in the above-described aspect, the heat transport direction of the first heat transport section and the heat transport direction of the second heat transport section differ from each other, the extending direction of the first heat transport section and the extending direction of the second heat transport section differ from each other.

(14) Further, in an aspect in which the container of the first heat transport section is of a planar type, and the container of the second heat transport section is of a planar type including the integral internal space, the heat transport device is configured as a vapor chamber.

(15) Since in the aspect of the heat transport device of the present disclosure, the capillary force of the first wick structure provided in the internal space of the first heat transport section and extending to the condensation section from the evaporation section is larger than the capillary force of the reflux promoting body provided on the second wick structure formed on the inner surface of the second heat transport section, and the capillary force of the reflux promoting body is larger than the capillary force of the second wick structure, or is equal to the capillary force of the second wick structure, the second wick structure and the reflux promoting body are provided on the inner surface of the second heat transport section, and the capillary force of the first wick structure provided in the first heat transport section including the evaporation section is in a mode of being relatively large. Consequently, according to the aspect of the heat transport device of the present disclosure, a reflux of the working fluid in a liquid phase to the direction of the connection portion with the first heat transport section from the tip end portion of the second heat transport section, and a reflux to the evaporation section from the condensation section of the first heat transport section are facilitated even under usage conditions in which the heating element with a high heat generation amount is thermally connected, or usage conditions in which a flow rate of cooling air is increased. Further, according to the aspect of the heat transport device of the present disclosure, the second wick structure and the reflux promoting body that are provided on the inner surface of the second heat transport section both have relatively small capillary forces, and therefore it is possible to prevent the working fluid in a liquid phase from being retained and accumulating in the second heat transport section. Further, according to the aspect of the heat transport device of the present disclosure, the working fluid in a liquid phase that flows back to the direction of the connection portion with the first heat transport section from the tip end portion of the second heat transport section in the second wick structure is protected from the flow of the working fluid in a gas phase that is a counterflow, by the reflux promoting body, and further, the reflux promoting body also

contributes to the reflux of the working fluid in a liquid phase to the direction of the connection portion with the first heat transport section from the tip end portion of the second heat transport section. From the above, according to the aspect of the heat transport device of the present disclosure, it is possible to prevent the working fluid in the evaporation section from drying out even under the usage conditions in which the heating element with a high heat generation amount is thermally connected, or the usage conditions in which the flow rate of cooling air is increased.

(16) According to the aspect of the heat transport device of the present disclosure, the first wick structure and the reflux promoting body are connected via the connection member having a capillary force, and thereby the reflux of the working fluid in a liquid phase to the first heat transport section from the second heat transport section is further facilitated.

(17) According to the aspect of the heat transport device of the present disclosure, the capillary force of the first wick structure is larger than the capillary force of the connection member, and the capillary force of the connection member is larger than the capillary force of the reflux promoting body, or is equal to the capillary force of the reflux promoting body, whereby the capillary force increases as the working fluid in a liquid phase flows to the first heat transport section from the second heat transport section, so that it is possible to prevent the working fluid in the evaporation section from drying out more reliably even under the usage conditions in which the heating element with a high heat generation amount is thermally connected, or the usage conditions in which the flow rate of the cooling air is increased.

(18) According to the aspect of the heat transport device of the present disclosure, in the section in the orthogonal direction to the heat transport direction of the second heat transport section, the reflux promoting body covers the region of 5% or more of the peripheral length of the section, and thereby the reflux promoting body further promotes a reflux of the working fluid in a liquid phase to the direction of the connection portion with the first heat transport section from the tip end portion of the second heat transport section.

(19) According to the aspect of the heat transport device of the present disclosure, the reflux promoting body includes the plurality of reflux promoting units, and the capillary forces of the plurality of reflux promoting units are different from one another, whereby the capillary force of the reflux promoting body can be increased as progress from the tip end portion of the second heat transport section towards the direction of the connection portion with the first heat transport section, so that a reflux of the working fluid in a liquid phase to the direction of the connection section with the first heat transport section from the tip end portion of the second heat transport section is further facilitated.

(20) According to the aspect of the heat transport device of the present disclosure, the plurality of second heat transport sections are provided and extend in the plurality of directions from the first heat transport section, and thereby the heat that is transported to the second heat transport section from the first heat transport section is transported in the plurality of directions different from the extending direction (heat transport direction) of the first heat transport section, so that it is possible to prevent increase in size of the heat transport device and, as a result, it is possible to achieve space saving.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a sectional plan view explaining an outline of an inside of a heat transport device according to a first embodiment of the present disclosure;

(2) FIG. 2 is a sectional front view explaining an outline of an inside of a second heat transport section of the heat transport device according to the first embodiment of the present disclosure;

(3) FIG. 3 is a sectional side view explaining the outline of the inside of the second heat transport

section of the heat transport device according to the first embodiment of the present disclosure;

(4) FIG. 4 is an explanatory view explaining a flow of a working fluid of the inside of the second heat transport section of the heat transport device according to the first embodiment of the present disclosure;

(5) FIG. 5 is a sectional plan view explaining an outline of an inside of a heat transport device according to a second embodiment of the present disclosure;

(6) FIG. 6 is a sectional front view explaining an outline of an inside of a second heat transport section of the heat transport device according to the second embodiment of the present disclosure;

(7) FIG. 7 is a sectional side view explaining the outline of the inside of the second heat transport section of the heat transport device according to the second embodiment of the present disclosure;

(8) FIG. 8 is a side view of a heat transport device according to a third embodiment of the present disclosure; and

(9) FIG. 9 is a sectional front view explaining an outline of an inside of a second heat transport section of a heat transport device according to a fourth embodiment of the present disclosure.

DETAILED DESCRIPTION

(10) Hereinafter, a heat transport device according to each of embodiments of the present disclosure will be described with use of the drawings. First, a heat transport device according to a first embodiment of the present disclosure will be described. Note that FIG. 1 is a sectional plan view explaining an outline of an inside of the heat transport device according to the first embodiment of the present disclosure. FIG. 2 is a sectional front view explaining an outline of an inside of a second transport section of the heat transport device according to the first embodiment of the present disclosure. FIG. 3 is a sectional side view explaining the outline of the inside of the second heat transport section of the heat transport device according to the first embodiment of the present disclosure. FIG. 4 is an explanatory view explaining a flow of a working fluid in the inside of the second heat transport section of the heat transport device according to the first embodiment of the present disclosure.

(11) As shown in FIG. 1, a heat transport device **1** according to the first embodiment of the present disclosure includes a first heat transport section **10** including an evaporation section **11** thermally connected to a heating element **100**, and a second heat transport section **20** connected at a condensation section **12** of the first heat transport section **10**, and having heat transport directions **D2** and **D3** that are different from a heat transport direction **D1** of the first heat transport section **10**. The second heat transport section **20** is connected to the first heat transport section **10** at the condensation section **12** of the first heat transport section **10**, and thereby, a connection portion **30** of the first heat transport section **10** and the second heat transport section **20** is formed. Further, an internal space of the first heat transport section **10** communicates with an internal space of the second heat transport section **20**. In the heat transport device **1**, the first heat transport section **10** has the integral internal space that provides communication from the evaporation section **11** to the connection portion **30** with the second heat transport section **20**, and has a working fluid enclosed in the internal space.

(12) The first heat transport section **10** has a container **19** having a hollow cavity portion **13**, and a working fluid flowing through the cavity portion **13** (not illustrated in FIG. 1). The container **19** is formed by overlapping one plate-shaped body **40** positioned on an installation surface side of the heat transport device **1** and another plate-shaped body (not illustrated in FIG. 1) facing the one plate-shaped body **40** with each other.

(13) The one plate-shaped body **40** is in a plate shape having side walls provided upright from a plane portion at edge portions of the plane portion. The other plate-shaped body is also in a plate shape having side walls provided upright from a plane portion at edge portions of the plane portion. Accordingly, the one plate-shaped body **40** and the other plate-shaped body are each in a recessed shape. By overlapping the one plate-shaped body **40** and the other plate-shaped body with each other in a state in which the recessed shapes face each other, the cavity portion **13** of the container

19 is formed. Accordingly, in the heat transport device **1**, a shape of the container **19** of the first heat transport section **10** is a planar shape. The cavity portion **13** of the container **19** is an internal space sealed to an external environment and is decompressed by degassing.

(14) Of an outer surface of the container **19**, a part to which the heating element **100** that is an object to be cooled is thermally connected functions as the evaporation section **11**. The heating element **100** is thermally connected to the container **19** of the first heat transport section **10**, and thereby the heating element **100** is cooled. In the first heat transport section **10**, the heating element **100** is thermally connected to one end in an extending direction that is the heat transport direction **D1** of the first heat transport section **10**. Accordingly, the evaporation section **11** is formed at the one end in the extending direction of the first heat transport section **10**.

(15) The first heat transport section **10** extends in a predetermined direction from a position of the heating element **100**, and the second heat transport section **20** is connected to another end facing the one end in the extending direction of the first heat transport section **10**. The other end of the first heat transport section **10** to which the second heat transport section **20** is connected functions as the condensation section **12** of the first heat transport section **10**.

(16) In the first heat transport section **10**, a middle portion that is positioned between the evaporation section **11** positioned at the one end of the container **19** and the condensation section **12** positioned at the other end of the container **19** functions as a heat-insulating portion **15**. Heat that is transmitted to the evaporation section **11** from the heating element **100** is transported to the condensation section **12** from the evaporation section **11** via the heat-insulating portion **15**.

(17) As shown in FIG. 1, in the cavity portion **13** of the first heat transport section **10**, a first wick structure **14** having a capillary force is housed. The first wick structure **14** is provided in the internal space of the first heat transport section **10**, and extends to the condensation section **12** from the evaporation section **11**. The first wick structure **14** extends to the condensation section **12** from the evaporation section **11** of the container **19** on a substantially entire surface in a width direction of the first heat transport section **10**.

(18) As the first wick structure **14**, it is possible to cite, for example, a sintered body of metal powder such as copper powder, metal mesh formed of a metal wire, nonwoven cloth, metal fiber and the like, though the first wick structure **14** is not particularly limited. In the first heat transport section **10**, as the first wick structure **14**, a sintered body of metal powder is used. Further, in the first heat transport section **10**, grooves (a plurality of thin grooves, not illustrated) may be further provided on an inner surface of the container **19**. In the cavity portion **13**, a part in which the first wick structure **14** is not provided functions as a vapor flow path **16** through which the working fluid in a gas phase flows. The vapor flow path **16** extends to the condensation section **12** from the evaporation section **11** of the container **19** correspondingly to that the first wick structure **14** extends to the condensation section **12** from the evaporation section **11** of the container **19**. The first heat transport section **10** transports heat of the heating element **100** received in the evaporation section **11** to the condensation section **12** from the evaporation section **11**, by heat transport properties due to operation of the working fluid.

(19) As shown in FIG. 1, the second heat transport section **20** the internal space of which communicates with the cavity portion **13** of the container **19** of the first heat transport section **10** is provided on a side surface of the condensation section **12** of the first heat transport section **10**. The second heat transport section **20** includes a container **29** having a hollow cavity portion **23**. The cavity portion **23** of the second heat transport section **20** communicates with the cavity portion **13** of the first heat transport section **10** via the connection portion **30**.

(20) The container **29** is formed by overlapping one plate-shaped body **41** positioned on an installation surface side of the heat transport device **1** and another plate-shaped body (not illustrated in FIG. 1) facing the one plate-shaped body **41** with each other. The one plate-shaped body **41** is in a plate shape having side walls provided upright from a plane portion at edge portions of the plane portion. The other plate-shaped body is also in a plate shape having side walls provided upright

from a plane portion at edge portions of the plane portion. Accordingly, the one plate-shaped body **41** and the other plate-shaped body are each in a recessed shape. By overlapping the one plate-shaped body **41** and the other plate-shaped body with each other in a state in which the recessed shapes face each other, the cavity portion **23** of the container **29** is formed. Accordingly, in the heat transport device **1**, a shape of the container **29** of the second heat transport section **20** is a planar shape.

(21) The cavity portion **23** of the container **29** is an internal space sealed to the external environment of the heat transport device **1** and is decompressed by degassing. The working fluid is flowable between the cavity portion **13** of the container **19** of the first heat transport section **10** and the cavity portion **23** of the container **29** of the second heat transport section **20**. From the above, the working fluid flowing through the cavity portion **13** of the first heat transport section **10** is enclosed in a space from the cavity portion **13** of the first heat transport section **10** to the cavity portion **23** of the second heat transport section **20**.

(22) A shape of the second heat transport section **20** is not particularly limited, and in the heat transport device **1**, the container **29** of the second heat transport section **20** is a planar shape having the integrated internal space.

(23) The second heat transport section **20** extends in a substantially orthogonal direction to the heat transport direction of the first heat transport section **10**, along a planar direction of the condensation section **12** of the first heat transport section **10**. Accordingly, the second heat transport section **20** may be installed along a direction orthogonal to a gravity direction. In the heat transport device **1**, the extending direction of the second heat transport section **20** is not parallel with the extending direction (heat transport direction) of the first heat transport section **10**, so that the heat transported from the first heat transport section **10** is transported in a direction different from the extending direction of the first heat transport section **10** by the second heat transport section **20**. Accordingly, it is possible to prevent increase in size of the heat transport device **1** in the heat transport direction of the first heat transport section **10**, and therefore it is possible to achieve space saving of the heat transport device **1**.

(24) A plurality of second heat transport sections **20** are provided, and extend in a plurality of directions from the first heat transport section **10**. In the heat transport device **1**, two second heat transport sections **20** are provided, and the second heat transport sections **20** extend in both left and right directions with the first heat transport section **10** as a center. Since the plurality of second heat transport sections **20** extend in a plurality of directions (two directions in the heat transport device **1**) from the first heat transport section **10**, the heat that is transported to the second heat transport sections **20** from the first heat transport section **10** diverges and is transported in a plurality of directions (two directions in the heat transport device **1**) different from the extending direction of the first heat transport section **10**. Accordingly, it is possible to prevent increase in size of the heat transport device **1** in the extending direction of the first heat transport section **10** more reliably.

(25) As shown in FIGS. **1** to **3**, a second wick structure **24** extending along the heat transport direction of the second heat transport section **20** is provided on an inner surface of the second heat transport section **20**. The second wick structure **24** has a function of causing the working fluid in a liquid phase to flow back in a direction of the connection portion **30** with the first heat transport section **10** from a tip end portion **33** of the second heat transport section **20**.

(26) The second wick structure **24** is a wick structure having a capillary force with a structure different from a structure of the first wick structure **14** housed in the container **19**. In the heat transport device **1**, as the second wick structure **24**, grooves (a plurality of thin grooves) are formed on an inner surface of the second heat transport section **20**. From the above, in the heat transport device **1**, the second wick structure **24** is a different type of wick structure from the first wick structure **14**. The grooves that are the second wick structure **24** extend along the heat transport direction of the second heat transport section **20**. In the heat transport device **1**, the second wick structure **24** extends from the tip end portion **33** of the second heat transport section **20** to the

connection portion **30** with the first heat transport section **10**.

(27) Further, in the second heat transport section **20**, a reflux promoting body **34** that extends along the heat transport direction of the second heat transport section **20** and has a capillary force is provided on the second wick structure **24**. The reflux promoting body **34** has a function of promoting the working fluid in a liquid phase flowing back to a direction of the connection portion **30** with the first heat transport section **10** from the tip end portion **33** of the second heat transport section **20**. The reflux promoting body **34** is provided in a mode of being in contact with the second wick structure **24**. In the heat transport device **1**, the reflux promoting body **34** extends from the tip end portion **33** of the second heat transport section **20** to a vicinity of the connection portion **30** with the first heat transport section **10**.

(28) The reflux promoting body **34** is a wick member that has a capillary force with a structure different from structures of the first wick structure **14** housed in the container **19** and the second wick structure **24** housed in the container **29**. As the reflux promoting body **34**, it is possible to cite, for example, a sintered body of metal powder such as copper powder, metal mesh formed of a metal wire, nonwoven fabric, metal fiber and the like, though the reflux promoting body **34** is not particularly limited. In the heat transport device **1**, the entire reflux promoting body **34** is formed of a same member, and the entire reflux promoting body **34** has a substantially same capillary force. In the heat transport device **1**, a sheet-shaped metal mesh is used as the reflux promoting body **34**. From the above, in the heat transport device **1**, the reflux promoting body **34** is a wick member of a different type from the first wick structure **14** and the second wick structure **24**. In the cavity portion **23** of the second heat transport section **20**, a part in which the reflux promoting body **34** and the second wick structure **24** are not provided functions as a vapor flow path **26** through which the working fluid in a gas phase flows. The vapor flow path **26** extends from the connection portion **30** with the first heat transport section **10** to the tip end portion **33** of the container **29** of the second heat transport section **20**. The second heat transport section **20** transports the heat of the heating element **100** received from the first heat transport section **10** to a direction of the tip end portion **33** of the second heat transport section **20** from the connection portion **30** with the first heat transport section **10** by the heat transport properties due to the operation of the working fluid.

(29) Though an installation position of the reflux promoting body **34** is not particularly limited as long as the installation position is on the inner surface of the container **29** of the second heat transport section **20**, it is preferable that, in a section in an orthogonal direction to the heat transport direction of the second heat transport section **20**, the reflux promoting body **34** covers a region of 5% or more of a peripheral length of the section, from a viewpoint that the working fluid in a liquid phase can flow back more smoothly to the direction of the connection portion **30** with the first heat transport section **10** from the tip end portion **33** of the second heat transport section **20**, by further promoting the reflux of the working fluid in a liquid phase to the direction of the connection portion **30** with the first heat transport section **10** from the tip end portion **33** of the second heat transport section **20**. Further, when the second heat transport section **20** is installed so that the extending direction of the second heat transport section **20** is substantially orthogonal to the gravity direction, it is preferable that the reflux promoting body **34** covers a region of 25% or more and 50% or less of the peripheral length of the aforementioned section, from a viewpoint of reliably securing the vapor flow path **26** while further promoting the reflux of the working fluid in a liquid phase to the direction of the connection portion **30** with the first heat transport section **10** from the tip end portion **33** of the second heat transport section **20**. Further, when the second heat transport section **20** is installed so that the tip end portion **33** of the second heat transport section **20** is lower in the gravity direction than the connection portion **30** with the first heat transport section **10**, it is preferable that the reflux promoting body **34** covers a region of 50% or more and 100% or less of the peripheral length of the aforementioned section, from a viewpoint of reliably promoting reflux of the working fluid in a liquid phase to the direction of the connection portion **30** from the tip end portion **33** of the second heat transport section **20**.

- (30) As shown in FIGS. 2 and 3, in the heat transport device 1, of the inner surface of the container 29 formed by overlapping the one plate-shaped body 41 and the other plate-shaped body 43 facing the one plate-shaped body 41 with each other, the reflux promoting body 34 is provided on the one plate-shaped body 41 that is the plate-shaped body positioned lower in the gravity direction, and is not provided on the other plate-shaped body 43.
- (31) In the heat transport device 1, a capillary force of the first wick structure 14 is in a mode of being larger than the capillary force of the reflux promoting body 34. Further, the capillary force of the reflux promoting body 34 is in a mode of being larger than the capillary force of the second wick structure 24, or being equal to the capillary force of the second wick structure 24.
- (32) As shown in FIGS. 1 and 2, the first wick structure 14 provided in the first heat transport section 10, and the reflux promoting body 34 provided in the second heat transport section 20 may be connected via a connection member 35 having a capillary force. The connection member 35 is positioned at the connection portion 30 of the first heat transport section 10 and the second heat transport section 20 and a vicinity of the connection portion 30.
- (33) The connection member 35 is a wick member that has a capillary force with a structure different from the structures of the first wick structure 14 housed in the container 19, and the second wick structure 24 and the reflux promoting body 34 that are housed in the container 29. As the connection member 35, it is possible to cite, for example, metal mesh, a braided body of a metal wire, metal fiber and the like. In the heat transport device 1, the connection member 35 may be a wick member of a different type from the first wick structure 14, the second wick structure 24 and the reflux promoting body 34.
- (34) In the heat transport device 1, the capillary force of the first wick structure 14 is in a mode of being larger than the capillary force of the connection member 35. Further, the capillary force of the connection member 35 is in a mode of being larger than the capillary force of the reflux promoting body 34, or substantially equal to the capillary force of the reflux promoting body 34.
- (35) The working fluid that changes in phase from a gas phase to a liquid phase inside of the second heat transport section 20 flows back to the direction of the connection portion 30 from the tip end portion 33 of the second heat transport section 20, inside of the second wick structure 24 and inside of the reflux promoting body 34 by the capillary force of the second wick structure 24 and the capillary force of the reflux promoting body 34. The working fluid in a liquid phase that flows back to the vicinity of the connection portion 30 flows to one end of the connection member 35 from the second wick structure 24 and the reflux promoting body 34. The working fluid in a liquid phase that flows to the one end of the connection member 35 flows to another end from the one end in the connection member 35 and flows back to the first wick structure 14 of the first heat transport section 10 from the other end of the connection member 35.
- (36) Accordingly, by providing the connection member 35, the reflux of the working fluid in a liquid phase to the first heat transport section 10 from the second heat transport section 20 is further facilitated. From the above, flowing performance of the working fluid in a liquid phase between the second heat transport section 20 and the first heat transport section 10 is improved by the connection member 35, and therefore the heat transport properties of the heat transport device 1 are improved.
- (37) Note that when the connection member 35 is not provided, it is preferable to adopt a structure in which the reflux promoting body 34 contacts the first wick structure 14 in the vicinity of the connection portion 30, from a viewpoint of improving the flowing performance of the working fluid in a liquid phase between the second heat transport section 20 and the first heat transport section 10.
- (38) As the materials of the container 19 and the container 29, it is possible to cite, for example, copper, a copper alloy, aluminum, an aluminum alloy, nickel, a nickel alloy, stainless steel, titanium, a titanium alloy and the like. The working fluid to be enclosed in the heat transport device 1 is properly selectable according to compatibility with the materials of the container 19 and the

container **29**, and it is possible to cite, for example, water, fluorocarbons, hydrofluoroether (HFE), cyclopentane, ethylene glycol, mixtures of them, and the like.

(39) Thicknesses of the container **19** and the container **29** are properly selectable based on mechanical strength, weight and the like, and it is possible to cite, for example, 0.5 to 3 mm. Widths of the container **19** and the container **29** are properly selectable according to usage conditions of the heat transport device **1** and the like, and it is possible to cite, for example, 4 to 20 mm.

(40) Next, a mechanism of a heat transport function of the heat transport device **1** will be described with use of FIGS. **1** to **3**. First, the heating element **100** that is the object to be cooled is thermally connected to one end of the container **19** of the first heat transport section **10** to cause the one end of the container **19** to function as the evaporation section **11**. When the one end of the container **19** receives heat from the heating element **100**, the heat is transmitted to the working fluid in a liquid phase of the cavity portion **13** at the one end of the container **19**, and the working fluid in a liquid phase changes in phase to the working fluid in a gas phase in the cavity portion **13** at the one end of the container **19**. The working fluid in a gas phase flows in the vapor flow path **16** from the one end of the container **19** to the other end that is the condensation section **12**. The working fluid in a gas phase flows to the other end from the one end of the container **19**, and thereby the first heat transport section **10** transports heat to the other end from the one end of the first heat transport section **10**. When a heat exchange section (not illustrated) is thermally connected at the other end of the container **19**, part of the working fluid in a gas phase that flows to the other end of the container **19** releases latent heat to change in phase to the working fluid in a liquid phase by a heat exchange action of the heat exchange section, and the released latent heat is transmitted to the heat exchange section. The heat that is transmitted to the heat exchange section is released to the external environment of the heat transport device **1** via the heat exchange section. The working fluid changing in phase to a liquid phase at the other end of the container **19** flows back to the one end from the other end of the container **19** by the capillary force of the first wick structure **14**.

(41) Further, since the cavity portion **13** of the first heat transport section **10** and the cavity portion **23** of the second heat transport section **20** communicate with each other, of the working fluid that changes in phase to a gas phase from a liquid phase in the evaporation section **11** of the first heat transport section **10**, the working fluid that does not change in phase to a liquid phase in the condensation section **12** of the first heat transport section **10** flows into the cavity portion **23** of the second heat transport section **20** from the cavity portion **13** of the first heat transport section **10** via the connection portion **30**. The working fluid in a gas phase that has flowed into the cavity portion **23** of the second heat transport section **20** from the cavity portion **13** of the first heat transport section **10** flows through the vapor flow path **26** in the direction of the tip end portion **33** of the second heat transport section **20** from the connection portion **30**. The working fluid in a gas phase flows through the vapor flow path **26** to the direction of the tip end portion **33** of the second heat transport section **20** from the connection portion **30**, and thereby the second heat transport section **20** transports heat to the direction of the tip end portion **33** from the connection portion **30**. When the heat exchange section (not illustrated) is thermally connected to the container **29** of the second heat transport section **20**, the working fluid in a gas phase that has flowed into the cavity portion **23** of the second heat transport section **20** releases latent heat inside of the second heat transport section **20** and changes in phase to the working fluid in a liquid phase by the heat exchange action of the heat exchange section. The latent heat released to inside of the second heat transport section **20** is transmitted to the heat exchange section thermally connected to the second heat transport section **20**. The heat that is transmitted to the heat exchange section is released to the external environment of the heat transport device **1** via the heat exchange section. The working fluid that changes in phase to a liquid phase from a gas phase inside of the second heat transport section **20** flows back to the direction of the connection portion **30** from the tip end portion **33** of the second heat transport section **20** by the capillary forces of the second wick structure **24** and the reflux

promoting body **34** provided on the inner surface of the second heat transport section **20**. The working fluid in a liquid phase that flows back to the vicinity of the connection portion **30** from the tip end portion **33** of the second heat transport section **20** flows back to the first wick structure **14** at the other end (condensation section **12**) of the container **19** via the connection member **35**. The working fluid in a liquid phase that has flowed back to the first wick structure **14** flows back to the one end (evaporation section **11**) of the container **19** from the other end (condensation section **12**) of the container **19** by the capillary force of the first wick structure **14**.

(42) In the heat transport device **1**, the capillary force of the first wick structure **14** provided in the cavity portion **13** of the first heat transport section **10** and extending to the condensation section **12** from the evaporation section **11** is larger than the capillary force of the reflux promoting body **34** provided on the second wick structure **24** formed on the inner surface of the second heat transport section **20**, and the capillary force of the reflux promoting body **34** is larger than the capillary force of the second wick structure **24** or the capillary force of the reflux promoting body **34** is substantially equal to the capillary force of the second wick structure **24**. Therefore, the second wick structure **24** and the reflux promoting body **34** are provided on the inner surface of the second heat transport section **20**, and the capillary force of the first wick structure **14** provided in the first heat transport section **10** including the evaporation section **11** is in a mode of being relatively large. Accordingly, in the heat transport device **1**, even under the usage conditions in which the heating element **100** with a high heat generation amount is thermally connected, or usage conditions in which the flow rate of the cooling air is increased, the reflux of the working fluid in a liquid phase to the direction of the connection portion **30** with the first heat transport section **10** from the tip end portion **33** of the second heat transport section **20**, and the reflux of the working fluid in a liquid phase to the evaporation section **11** from the condensation section **12** of the first heat transport section **10** are facilitated. Further, in the heat transport device **1**, the second wick structure **24** and the reflux promoting body **34** that are provided on the inner surface of the second heat transport section **20** both have relatively small capillary forces, and therefore it is possible to prevent the working fluid in a liquid phase from being retained and accumulating in the second heat transport section **20**.

(43) Further, as shown in FIG. **4**, in the heat transport device **1**, a working fluid **F1** in a liquid phase that flows back to the direction of the connection portion **30** with the first heat transport section **10** from the tip end portion **33** of the second heat transport section **20** in the second wick structure **24** is protected from a flow of a working fluid **G** in a gas phase that is a counterflow, by the reflux promoting body **34**, and further, the reflux promoting body **34** also contributes to reflux of a working fluid **F2** in a liquid phase to the direction of the connection portion **30** with the first heat transport section **10** from the tip end portion **33** of the second heat transport section **20**.

Furthermore, an interface portion **S** between the second wick structure **24** and the reflux promoting body **34** also contributes to reflux of a working fluid **F3** in a liquid phase to the direction of the connection portion **30** with the first heat transport section **10** from the tip end portion **33** of the second heat transport section **20**. From the above, in the heat transport device **1**, it is possible to prevent the working fluid in the evaporation section **11** from drying out even under the usage conditions in which the heating element **100** with a high heat generation amount is thermally connected or usage conditions in which the flow rate of the cooling air is increased.

(44) Further, since in the heat transport device **1**, the capillary force of the first wick structure **14** is larger than the capillary force of the connection member **35**, and the capillary force of the connection member **35** is larger than the capillary force of the reflux promoting body **34**, or the capillary force of the connection member **35** is substantially equal to the capillary force of the reflux promoting body **34**, the capillary force is in a mode of becoming larger as the working fluid in a liquid phase flows to the first heat transport section **10** from the second heat transport section **20**, so that it is possible to prevent the working fluid in the evaporation section **11** from drying out more reliably even under the usage conditions in which the heating element **100** with a high heat

generation amount is thermally connected, or the usage conditions in which the flow rate of the cooling air is increased.

(45) Next, a heat transport device according to a second embodiment of the present disclosure will be described with use of the drawings. Note that the heat transport device according to the second embodiment shares a main part with the heat transport device according to the first embodiment, and therefore, same components as the components of the heat transport device according to the first embodiment will be described by using the same reference signs. Note that FIG. 5 is a sectional plan view explaining an outline of an inside of the heat transport device according to the second embodiment of the present disclosure. FIG. 6 is a sectional front view explaining an outline of an inside of a second heat transport section of the heat transport device according to the second embodiment of the present disclosure. FIG. 7 is a sectional side view explaining the outline of the inside of the second heat transport section of the heat transport device according to the second embodiment of the present disclosure.

(46) In the heat transport device **1** according to the first embodiment, the second heat transport section **20** is a planar type of container including the integral internal space, but as shown in FIG. 5, in a heat transport device **2** according to the second embodiment, a second heat transport section **20** is a plurality of tube bodies **51**, **51**, **51** . . . instead of the planar type of container including the integral internal space.

(47) The second heat transport section **20** includes a plurality of tube bodies **51** each having a hollow cavity portion **23**. The plurality of tube bodies **51**, **51**, **51** . . . are disposed in parallel along a radial direction of the tube body **51**, and the one second heat transport section **20** is formed. A longitudinal direction of the tube body **51** is a heat transport direction of the tube body **51**, and thus, a heat transport direction of the second heat transport section **20**.

(48) An end portion (hereinafter, may be referred to as “base portion”) **32**, on a first heat transport section **10** side, of the tube body **51** is opened, and an end portion opposite to the base portion **32** (that is, a tip end portion **33**) is closed. The cavity portion **23** of the tube body **51** forming the second heat transport section **20** communicates with a cavity portion **13** of the first heat transport section **10** via a connection portion **30**. The cavity portion **23** of the tube body **51** is an internal space sealed to an external environment of the heat transport device **2** and is decompressed by degassing. A working fluid is flowable between the cavity portion **13** of the first heat transport section **10** and the cavity portion **23** of the tube body **51**.

(49) A through-hole (not illustrated) for mounting the tube body **51** to the container **19** is formed in a side surface portion of the container **19** of the first heat transport section **10**. A shape and a size of the through-hole correspond to a shape and a size of the tube body **51**, and the base portion **32** of the tube body **51** is fitted into the through-hole of the container **19**, whereby the tube body **51** is connected to the first heat transport section **10**. Accordingly, the tube body **51** and the container **19** of the first heat transport section **10** are formed of separate members. Though a method for fixing the tube body **51** mounted to the container **19** is not particularly limited, it is possible to cite, for example, welding, soldering, brazing, and the like. The tube body **51** extends in a substantially orthogonal direction to the heat transport direction of the first heat transport section **10**, along a planar direction of a condensation section **12** of the first heat transport section **10**. Accordingly, in the heat transport device **2**, a longitudinal direction of the tube body **51** may be installed so that the longitudinal direction of the tube body **51** is along a direction orthogonal to the gravity direction.

(50) As shown in FIGS. 5 to 7, a second wick structure **24** extending along the heat transport direction of the tube body **51** is provided on an inner surface of each of the tube bodies **51**. The second wick structure **24** has a function of causing the working fluid in a liquid phase to a direction of the connection portion **30** with the first heat transport section **10** from the tip end portion **33** of the tube body **51**. Also in the heat transport device **2**, the second wick structure **24** extends to the connection portion **30** with the first heat transport section **10** from the tip end portion **33** of the tube body **51**. Also in the heat transport device **2**, grooves (a plurality of fine grooves) are formed on the

inner surface of the tube body **51** as the second wick structure **24**.

(51) Further, a reflux promoting body **34** having a capillary force, and extending along the heat transport direction of the second heat transport section **20** is provided on the second wick structure **24** in each of the tube bodies **51**. In the heat transport device **2**, the reflux promoting body **34** extends to a vicinity of the connection portion **30** with the first heat transport section **10** (that is, a vicinity of the base portion **32**) from a center portion **36** in the longitudinal direction of the tube body **51**. From the above, in the heat transport device **2**, in a region from the tip end portion **33** to the center portion **36** of the tube body **51**, the reflux promoting body **34** is not provided, and the second wick structure **24** is exposed. The reflux promoting body **34** is provided in a lower part in the gravity direction and is not provided in an upper part in the gravity direction of the inner surface of the tube body **51**. Note that also in the heat transport device **2**, the entire reflux promoting body **34** is formed of a same member and has a substantially same capillary force.

(52) Also in the heat transport device **2**, a first wick structure **14** provided in the first heat transport section **10**, and the reflux promoting body **34** provided in the tube body **51** may be connected via a connection member **35** having a capillary force.

(53) A shape of the tube body **51** is not particularly limited, a shape in the longitudinal direction is a straight-line shape, and a shape in an orthogonal direction (radial direction) to the longitudinal direction is an elliptical shape, in the heat transport device **2**. Further, each of the tube bodies **51** also has a substantially same shape and size. As the material of the tube body **51** that forms the second heat transport section **20**, it is possible to cite, for example, copper, a copper alloy, aluminum, an aluminum alloy, nickel, or a nickel alloy, stainless steel, titanium, a titanium alloy, and the like, like the material of the container **29** of the second heat transport section **20**.

(54) As shown in FIGS. **5** to **7**, in the heat transport device **2**, a plurality of first heat dissipation fins **61**, **61**, **61** . . . that are heat exchange sections are thermally connected to the tube body **51**. The plurality of first heat dissipation fins **61**, **61**, **61** . . . are disposed in parallel along the longitudinal direction of the tube body **51**. The tube body **51** is fitted into each of a plurality of through-holes formed in the first heat dissipation fin **61**, whereby the first heat dissipation fin **61** is mounted and fixed to a position of the tube body **51**, and the first heat dissipation fin **61** is thermally connected to the plurality of tube bodies **51**, **51**, **51** From the above, the heat transport device **2** is a heat sink in which the plurality of first heat dissipation fins **61**, **61**, **61** . . . are thermally connected to the second heat transport section **20**.

(55) Further, as shown in FIG. **6**, a plurality of second heat dissipation fins **62**, **62**, **62** . . . that are heat exchange sections are thermally connected to the condensation section **12** of the first heat transport section **10** (not illustrated for convenience of explanation in FIG. **5**). The plurality of second heat dissipation fins **62**, **62**, **62** . . . are disposed in parallel along a width direction of the first heat transport section **10**. The second heat dissipation fin **62** is mounted and fixed to the position of the first heat transport section **10**, and is thermally connected to the first heat transport section **10**. The first heat dissipation fin **61** and the second heat dissipation fin **62** are both thin flat-plate-shaped members.

(56) Also in the heat transport device **2**, a capillary force of the first wick structure **14** provided in the cavity portion **13** of the first heat transport section **10** and extending to the condensation section **12** from the evaporation section **11** is larger than the capillary force of the reflux promoting body **34** provided on the second wick structure **24** formed on the inner surface of the tube body **51** that is the second heat transport section **20**, the capillary force of the reflux promoting body **34** is larger than the capillary force of the second wick structure **24**, or the capillary force of the reflux promoting body **34** is substantially equal to the capillary force of the second wick structure **24**, so that the second wick structure **24** and the reflux promoting body **34** are provided on the inner surface of the tube body **51**, and the capillary force of the first wick structure **14** provided in the first heat transport section **10** including the evaporation section **11** is in a mode of being relatively large. Accordingly, also in the heat transport device **2**, a reflux of the working fluid in a liquid

phase to the direction of the connection portion **30** with the first heat transport section **10** from the tip end portion **33** of the tube body **51** and a reflux of the working fluid in a liquid phase to the evaporation section **11** from the condensation section **12** of the first heat transport section **10** are facilitated even under the usage conditions in which the heating element **100** with a high heat generation amount is thermally connected, or under the usage conditions in which a flow rate of the cooling air is increased. Furthermore, also in the heat transport device **2**, both the second wick structure **24** and the reflux promoting body **34** that are provided on the inner surface of the tube body **51** have relatively small capillary forces, and therefore it is possible to prevent the working fluid in a liquid phase from being retained and accumulating in the tube body **51**.

(57) Further, also in the heat transport device **2**, the working fluid in a liquid phase that flows back to the direction of the connection portion **30** with the first heat transport section **10** from the tip end portion **33** of the tube body **51** in the second wick structure **24** is protected from a flow of the working fluid in a gas phase that is a counterflow, by the reflux promoting body **34**, and further, the reflux promoting body **34** also contributes to the reflux of the working fluid in a liquid phase to the direction of the connection portion **30** with the first heat transport section **10** from the tip end portion **33** of the tube body **51**. Furthermore, an interface portion between the second wick structure **24** and the reflux promoting body **34** also contributes to the reflux of the working fluid in a liquid phase to the direction of the connection portion **30** with the first heat transport section **10** from the tip end portion **33** of the tube body **51**. From the above, also in the heat transport device **2**, it is possible to prevent the working fluid in the evaporation section **11** from drying out even in the usage conditions in which the heating element **100** with a high heat generation amount is thermally connected, or the usage conditions in which the flow rate of cooling air is increased.

(58) Next, a heat transport device according to a third embodiment of the present disclosure will be described with use of the drawings. Note that the heat transport device according to the third embodiment shares a main part with the heat transport devices according to the first and second embodiments, and therefore same components as the components of the heat transport devices according to the first and second embodiments will be described by using the same reference signs. Note that FIG. **8** is a side view of the heat transport device according to the third embodiment of the present disclosure.

(59) In the heat transport devices **1** and **2** of the first and second embodiments, the second heat transport section **20** extends in the substantially orthogonal direction to the heat transport direction of the first heat transport section **10** along the planar direction of the condensation section **12** of the first heat transport section **10**, but as shown in FIG. **8**, in a heat transport device **3** according to the third embodiment, a second heat transport section **20** extends in a perpendicular direction to a planar direction of a condensation section **12** of a first heat transport section **10**, and in a substantially orthogonal direction to a heat transport direction of the first heat transport section **10**.

(60) In the heat transport device **3**, the second heat transport section **20** is composed of a plurality of tube bodies **51**, **51**, **51** . . . , and the plurality of tube bodies **51**, **51**, **51** . . . forming the second heat transport section **20** extend downward in the gravity direction. In the heat transport device **3**, one second heat transport section **20** is provided. Since a tip end portion **33** of the second heat transport section **20** is installed below an evaporation section **11** of the first heat transport section **10** in the gravity direction, it is preferable that a reflux promoting body covers a region of 50% or more and 100% or less of a peripheral length in a section in an orthogonal direction to a heat transport direction of the tube body **51**, from a viewpoint of reliably promoting a reflux of the working fluid in a liquid phase to the first heat transport section **10** from the tip end portion **33** of the second heat transport section **20**.

(61) Also in the heat transport device **3**, a second wick structure and the reflux promoting body are provided on an inner surface of the tube body **51**, and a capillary force of a first wick structure provided in the first heat transport section **10** including the evaporation section **11** is in a mode of being relatively large, so that a reflux of the working fluid in a liquid phase to a direction of the

first heat transport section **10** from the tip end portion **33** of the tube body **51** and a reflux of the working fluid in a liquid phase to the evaporation section **11** from the condensation section **12** of the first heat transport section **10** are facilitated even under usage conditions in which a heating element **100** with a high heat generation amount is thermally connected or usage conditions in which a flow rate of cooling air is increased.

(62) Next, a heat transport device according to a fourth embodiment of the present disclosure will be described with use of the drawings. Note that the heat transport device according to the fourth embodiment shares a main part with the heat transport devices according to the first to third embodiments, and therefore same components as the components of the heat transport devices according to the first to third embodiments will be described by using the same reference signs. Note that FIG. **9** is a sectional front view explaining an outline of an inside of a second heat transport section of the heat transport device according to the fourth embodiment of the present disclosure.

(63) In each of the heat transport devices **1**, **2** and **3** according to the first to third embodiments, the entire reflux promoting body **34** is formed of the same member, and the entire reflux promoting body **34** has substantially the same capillary force, whereas as shown in FIG. **9**, in a heat transport device **4** according to the fourth embodiment, the reflux promoting body **34** has a structure in which the reflux promoting body **34** is composed of a plurality of reflux promoting units **34a**, **34a**, **34a** . . . , and capillary forces of the plurality of reflux promoting units **34a**, **34a**, **34a** . . . are different from one another. Note that, in FIG. **9**, the reflux promoting body **34** has three reflux promoting units **34a** for convenience of explanation, but the number of reflux promoting units **34a** to be installed is not particularly limited as long as the number is plural.

(64) A capillary force of each of the reflux promoting units **34a** is in a mode of being smaller than a capillary force of a first wick structure **14** correspondingly to that the capillary force of the first wick structure **14** is in a mode of being larger than a capillary force of the reflux promoting body **34**. Further, the capillary force of each of the reflux promoting units **34a** is also in a mode of being larger than a capillary force of a second wick structure **24**, or being substantially equal to the capillary force of the second wick structure **24** correspondingly to that the capillary force of the reflux promoting body **34** is in a mode of being larger than the capillary force of the second wick structure **24**, or substantially equal to the capillary force of the second wick structure **24**.

(65) The plurality of reflux promoting units **34a**, **34a**, **34a** . . . are disposed to be continuously connected along a heat transport direction of a second heat transport section **20**. In the heat transport device **4**, the plurality of reflux promoting units **34a**, **34a**, **34a** . . . are disposed to be in direct contact with one another. When the reflux promoting units **34a** are metal mesh, for example, it is possible to select a magnitude of a capillary force of each of the plurality of reflux promoting units **34a**, **34a**, **34a** . . . by adjusting a thickness of the metal mesh, and when the reflux promoting unit **34a** is a sintered body of metal powder, it is possible to select the magnitude of the capillary force by adjusting a porosity of the sintered body.

(66) The capillary forces of the plurality of reflux promoting units **34a**, **34a**, **34a** . . . are different from one another, and thereby it is possible to set the capillary force of the reflux promoting body **34** to a larger capillary force as progress from the tip end portion **33** of the second heat transport section **20** toward the first heat transport section **10**, so that the reflux of the working fluid in a liquid phase to the direction of the first heat transport section **10** from the tip end portion **33** of the second heat transport section **20** is further facilitated.

(67) Next, another embodiment of the heat transport device of the present disclosure will be described hereinafter. In each of the heat transport devices of the above-described first and second embodiments, the plurality of second heat transport sections are provided, but instead of this, the number of second heat transport sections may be one.

(68) Since the heat transport device of the present disclosure can prevent the working fluid in the evaporation section from drying out even under the usage conditions in which an input heat amount

is high and the flow rate of the cooling air is increased, the heat transport device has a high utility value in the field of cooling an electronic component with a high heat generation amount mounted on a circuit board, for example, an electronic component such as a central processing unit, for example.

Claims

1. A heat transport device, comprising: a first heat transport section including an evaporation section thermally connected to a heating element; and a second heat transport section connected at a condensation section of the first heat transport section, and having a heat transport direction different from a heat transport direction of the first heat transport section, the first heat transport section including an integral internal space that provides communication from the evaporation section to a connection portion with the second heat transport section and has a working fluid enclosed in the internal space, the internal space of the first heat transport section communicating with an internal space of the second heat transport section, the heat transport device comprising a first wick structure provided in the internal space of the first heat transport section, and extending from the evaporation section to the condensation section; a second wick structure provided on an inner surface of the second heat transport section, and extending along the heat transport direction of the second heat transport section; and a reflux promoting body in contact with the second wick structure, extending along the heat transport direction of the second heat transport section, and having a capillary force, wherein a capillary force of the first wick structure is larger than the capillary force of the reflux promoting body, and the capillary force of the reflux promoting body is larger than a capillary force of the second wick structure, or the capillary force of the reflux promoting body is equal to the capillary force of the second wick structure, and the reflux promoting body is located on the inner surface of the second heat transport section, the reflux promoting body is provided on a lower side in a gravity direction and is not provided on an upper side in the gravity direction.

2. A heat transport device, comprising: a first heat transport section including an evaporation section thermally connected to a heating element; and a second heat transport section connected at a condensation section of the first heat transport section, and having a heat transport direction different from a heat transport direction of the first heat transport section, the first heat transport section including an integral internal space that provides communication from the evaporation section to a connection portion with the second heat transport section and has a working fluid enclosed in the internal space, the internal space of the first heat transport section communicating with an internal space of the second heat transport section, the heat transport device comprising a first wick structure provided in the internal space of the first heat transport section, and extending from the evaporation section to the condensation section; a second wick structure provided on an inner surface of the second heat transport section, and extending along the heat transport direction of the second heat transport section; and a reflux promoting body in contact with the second wick structure, extending along the heat transport direction of the second heat transport section, and having a capillary force, wherein a capillary force of the first wick structure is larger than the capillary force of the reflux promoting body, and the capillary force of the reflux promoting body is larger than a capillary force of the second wick structure, or the capillary force of the reflux promoting body is equal to the capillary force of the second wick structure, and the reflux promoting body covers a region of 5% or more and 50% or less of a peripheral length of a section, the section being in an orthogonal direction to the heat transport direction of the second heat transport section.

3. A heat transport device, comprising: a first heat transport section including an evaporation section thermally connected to a heating element; and a second heat transport section connected at a condensation section of the first heat transport section, and having a heat transport direction

different from a heat transport direction of the first heat transport section, the first heat transport section including an integral internal space that provides communication from the evaporation section to a connection portion with the second heat transport section and has a working fluid enclosed in the internal space, the internal space of the first heat transport section communicating with an internal space of the second heat transport section, the heat transport device comprising a first wick structure provided in the internal space of the first heat transport section, and extending from the evaporation section to the condensation section; a second wick structure provided on an inner surface of the second heat transport section, and extending along the heat transport direction of the second heat transport section; and a reflux promoting body in contact with the second wick structure, extending along the heat transport direction of the second heat transport section, and having a capillary force, wherein a capillary force of the first wick structure is larger than the capillary force of the reflux promoting body, and the capillary force of the reflux promoting body is larger than a capillary force of the second wick structure, or the capillary force of the reflux promoting body is equal to the capillary force of the second wick structure, and the second heat transport section is installed so that a tip end portion of the second heat transport section is lower in a gravity direction than the connection portion with the first heat transport section.

4. The heat transport device according to claim 1, wherein the second heat transport section is installed so that an extending direction of the second heat transport section is an orthogonal direction to the gravity direction.

5. The heat transport device according to claim 1, wherein the first wick structure and the reflux promoting body are connected via a connection member having a capillary force.

6. The heat transport device according to claim 5, wherein a capillary force of the first wick structure is larger than the capillary force of the connection member, and the capillary force of the connection member is larger than the capillary force of the reflux promoting body, or the capillary force of the connection member is the same as the capillary force of the reflux promoting body.

7. The heat transport device according to claim 6, wherein the connection member is located at a connection portion between the first heat transport section and the second heat transport section.

8. The heat transport device according to claim 1, wherein the reflux promoting body includes a plurality of reflux accelerating portions, and capillary forces of the plurality of reflux accelerating portions are different from each other.

9. The heat transport device according to claim 1, wherein the container of the first heat transport section has a planar edge.

10. The heat transport device according to claim 1, wherein the container of the second heat transport section is of a planar type having an integrated internal space.

11. The heat transport device according to claim 1, wherein the second heat transport section includes a plurality of pipe bodies.

12. The heat transport device according to claim 11, wherein the plurality of pipe bodies extends in a plurality of directions from the first heat transport section.

13. The heat transport device according to claim 11, wherein the plurality of pipe bodies extends downward in the direction of gravity.

14. The heat transport device according to claim 1, wherein the heat transport device is a heat sink in which a plurality of heat radiation fins is thermally connected to the second heat transport section.

15. The heat transport device according to claim 1, wherein the reflux promoting body is a wick member having a capillary force and has a structure different from structures of the first wick structure and the second wick structure.

16. The heat transport device according to claim 2, wherein the reflux promoting body is a wick member having a capillary force and has a structure different from structures of the first wick structure and the second wick structure.

17. The heat transport device according to claim 3, wherein the reflux promoting body is a wick

member having a capillary force and has a structure different from structures of the first wick structure and the second wick structure.
